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Efficient water quality prediction by synthesizing seven heavy metal parameters using deep neural network

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ABSTRACT

Water quality is primarily assessed by consumers based on its aesthetic acceptability, encompassing factors such as taste, odor, and appearance. The seven heavy metal parameters - magnesium, sulfate, potassium, sodium, calcium, total hardness, and chloride - significantly impact the aesthetic acceptability of water, playing crucial roles in influencing factors such as taste and odor. In this study, we present a deep learning-based approach to reconstruct these seven parameters from four parameters (temperature, electrical conductivity, potential of hydrogen, and total dissolved solids) in order to estimate the water quality index. This approach simplifies data collection, making it cost-effective without the need for expensive devices or traditional lab facilities. It also facilitates the assessment of sample quality by calculating the water quality index using these seven synthesized parameters in conjunction with recorded potential of hydrogen and total dissolved solids values. Accurate assessment of these parameters is essential for evaluating the suitability of water for various purposes, including consumption and industrial applications. Our proposed model is trained on a comprehensive dataset encompassing a wide range of water samples. Through extensive experimentation and evaluation, we demonstrate that our deep learning model effectively reconstructs the seven parameters with high accuracy and reliability. Our study's results demonstrate that utilizing reconstructed parameters achieves performance comparable to that of using actual recorded parameters for determining the quality of each sample. This approach offers a cost-effective and efficient solution for understanding water quality, enabling improved decision-making and management of water resources.

1. Introduction

The water quality index (WQI) is a critical tool used to assess and communicate the overall quality of water resources, providing valuable information for policymakers, water managers, and the general public. Different water quality indices can yield diverse outcomes for the same water sample [1]. Furthermore, the selection of parameters significantly influences the WQI results, emphasizing the importance of parameter choice in water quality assessment [2]. For instance, the range of parameters varies from four in the Canadian council of ministers of the environment (CCME) model [3], to twenty-six in the Bascaron model [4]. Nevertheless, even within a single model, various studies employed diverse parameter sets depending on data availability and expert perspectives. Therefore, the assessment of water quality using physico-chemical parameters is limited to the properties that have been recorded. If certain parameters in heavy metals or pesticides were not measured nor included in the calculation of the water quality index, the water could potentially contain dangerous levels of these substances, while the WQI could show no contamination [2].

There are various factors that impact the capability to record a comprehensive set of water quality parameters, such as limits of monitoring devices, the cost of advanced equipment, and the expenses involved in laboratory analysis [5,6]. Furthermore, since water-related data is complex due to unpredictable natural changes, human interference, and nonstationary features, it is necessary to use reliable methodologies to deal with the poor quality of data [7]. Soft sensors, also known as virtual sensors can provide real-time estimates of difficult-to-measure/expensive-to-measure variables and have been considered a

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valuable alternative for direct measuring parameters [8]. Machine learning models have been proposed as a reliable method to simulate soft sensors based on the available water-related data [8].

The existing literature on this topic primarily focuses on utilizing machine learning techniques to predict water quality parameters for multiple applications. For instance, forecasting the levels of dissolved oxygen (DO) and biochemical oxygen demand (BOD) is crucial for effectively monitoring the health of a river ecosystem, as these parameters offer valuable insights into the overall condition of the waterbody [9–11]. Similarly, monitoring total nitrogen (TN) and total phosphorus (TP) is important as their excessive levels can cause eutrophication and depleted oxygen levels, adversely affecting water quality and aquatic life [12,13]. Additionally, various studies monitored and artificially measured important water quality parameters in drinking water such as total dissolved solids (TDS), total hardness (TH), sulfate (SO_4), and chloride (Cl) [14–16]. These studies used different machine learning techniques and the number of input variables to synthesize water quality parameters. The input variables used for predicting water quality parameters include both physical parameters, such as temperature and turbidity, as well as chemical factors, like the TDS and TN. The most widely employed method for predicting water quality parameters is through the use of artificial neural networks (ANN) and its various adaptations [7]. For example, the ANNs are commonly utilized to forecast the DO parameter using various input sets, ranging from one to twenty-five [9,17–20]. Polynomial neural network (PNN) is used with 18 water quality parameters to estimate DO concentrations [21]. A cascade correlation neural network (CCNN) model with a different combination of five water quality parameters is used to predict DO concentrations [22]. Gated recurrent unit (GRU) in collaboration with particle swarm optimization (PSO) is utilized to predict DO parameter [9]. Deep matrix factorization (DMF) and deep neural network (DNN) are used to predict the BOD using twenty-six input parameters [23]. Support vector machine (SVM) and the ANN models are used to forecast the TN and TP based on eight parameters [24]. Certain studies employed all available water quality parameters to predict a single parameter, repeating this process for each parameter individually. For example, one study used extreme gradient boosting (XGBoost) and random forest (RF) with a set of five input parameters [25], whereas in another study, the SVM used a set of eight input parameters [16].

This paper introduces a comprehensive framework for evaluating water quality, focusing on its overall acceptability determined by factors like odor, taste, and appearance. Our approach employs four readily measurable parameters to estimate seven key heavy metal parameters and subsequently calculate the water quality index. In the realm of water quality research, prior works have delved into employing different machine learning techniques to forecast the water quality index utilizing existing parameters [26–29]. However, our study distinguishes itself by placing its primary emphasis on synthesizing seven distinct parameters, which we often refer to as *artificially measured*. Subsequently, our goal is to evaluate the WQI based on these synthesized parameters. The incorporation of water temperature (WT), electrical conductivity (EC), potential of hydrogen (pH), and total dissolved solids (TDS) in our machine learning model enables the accurate prediction of several water quality parameters, including magnesium (Mg), sulfate (SO_4), potassium (K), sodium (Na), calcium (Ca), total hardness (TH), and chloride (Cl). These four key parameters can be measured using basic field instruments such as thermometer, conductivity meter, pH meter, test strips, and TDS meter, which are widely available and relatively inexpensive. Therefore, this approach eliminates the need for expensive devices or access to laboratory facilities traditionally associated with measuring different parameters in water for water quality index assessment. Our framework harnesses the power of these parameters, offering a practical and cost-effective solution for monitoring water quality. This makes large-scale data collection on water quality parameters across different locations over time more accessible. This streamlined approach enhances accessibility and reduces the economic burden of water quality monitoring. This is particularly valuable in resource-limited settings such as third world countries where access to specialized equipment or trained personnel may be limited, ensuring that water quality assessment is feasible and accessible to a broader range of communities.

2. Materials and method

Our study introduces an innovative framework for predicting water quality, leveraging measurements of four easily and cost-effectively accessible parameters: temperature, EC, pH, and TDS. This framework comprises three key components:

Deep neural network framework: Initially, we employ seven distinct deep neural networks, each specialized in predicting one of the heavy metal parameters: Mg, SO_4, K, Na, TH, Cl, and Ca. These DNNs leverage four accessible parameters as input: EC, TDS, WT, and pH.

WQI computation: In the second phase, after the calculation of the water quality index for each sample using the synthesized parameters in combination with recorded TDS and pH values, the sample's quality label is determined based on this calculated WQI. Our evaluation relies on two water quality indices: the heavy metal evaluation index (HEI) [30,31] and the entropy-weighted heavy metal contamination index (EHCI) [32,33], offering a comprehensive assessment of each sample's quality.

Quantification: Given that our WQI calculation includes seven synthesized parameters and two recorded parameters (TDS and pH), we aim to quantify potential errors when compared to using the nine recorded parameters for WQI assessment. To achieve this, we calculate the WQI for all samples using the nine recorded parameters, utilizing the corresponding label for each sample as a *reference* point for quantifying this error.

In this paper, the term *classification* is used to describe the assignment of labels to each sample using HEI or EHCI methods (Table 1). Subsequently, we evaluate this performance by comparing it to the reference labels. We refer to the combined set of seven synthesized parameters and two recorded parameters as “combined parameters”, while only TDS and pH are referred to as “dual parameters”. We conducted a comparative analysis between these two approaches to demonstrate the effectiveness of using the seven synthesized parameters in calculating the WQI. In Fig. 1, a diagram illustrating the process is presented.

In addition to the technical aspects of water quality assessment, it is crucial to recognize the influence of various ions on the taste, odor, and appearance of water. Consumers lack the facilities to technically evaluate the safety of water, leading them to rely on its aesthetic acceptability as an indicator of quality. Taste, odor, and appearance are key factors that heavily influence their perception of water quality. As a result, any issues or doubts regarding these aspects can drive consumers to seek alternative water sources that appear more aesthetically appealing, even if they may pose potential risks to overall water quality [34]. The taste, odor, and appearance of water can be significantly affected by the presence of various ions such as Mg, SO_4, K, Na, TH, Cl, and Ca [34]. The impact of each of these parameters on water quality will be briefly described below:

Cl: Monitoring and maintaining the Cl levels in drinking water is important to prevent taste, odor, and quality issues that can arise when concentrations exceed 250 mg/L. Excessive chloride levels have the potential to contribute to the corrosion of pipes and plumbing fixtures, leading to increased levels of lead in the water supply [34].

EC: Water conductivity, influenced by dissolved ions, indirectly affects the taste, odor, and color of water. Higher concentrations of salts can cause a salty taste, while specific minerals may contribute to metallic or bitter flavors. The WHO does not provide any recommended permissible limit for electrical conductivity [34].

pH: Water with a low pH can exhibit corrosive properties, causing the leaching of metals such as lead, copper, and zinc from plumbing systems [34,35]. Ingesting water with elevated levels of these metals can potentially pose health risks. On the other hand, high pH levels can

Table 1
Statistical summary of water parameters with WHO permissible limits.

Indicator	Unit	Minimum	Maximum	Mean	Standard deviation	WHO limit
EC	$\mu S/cm$	50.00	49,500	1744.70	5223.35	–
pH	–	4.61	11.7	6.75	0.56	6.5–8.5
TDS	mg/L	37.00	36,500	1177.26	3791.72	1000
WT	°C	19.30	34	27.04	1.72	–
Cl	mg/L	1.30	17,700	388.28	1883.22	250
c	mg/L	0.90	2410	125.86	256.97	250
TH	mg/L	2.00	5460	390.20	616.55	500
Na	mg/L	2.53	9760	244.75	1074.55	200
K	mg/L	0.13	373	12.70	39.14	10
Mg	mg/L	0.10	993	42.68	116.15	250
Ca	mg/L	1.44	547	86.80	67.90	300

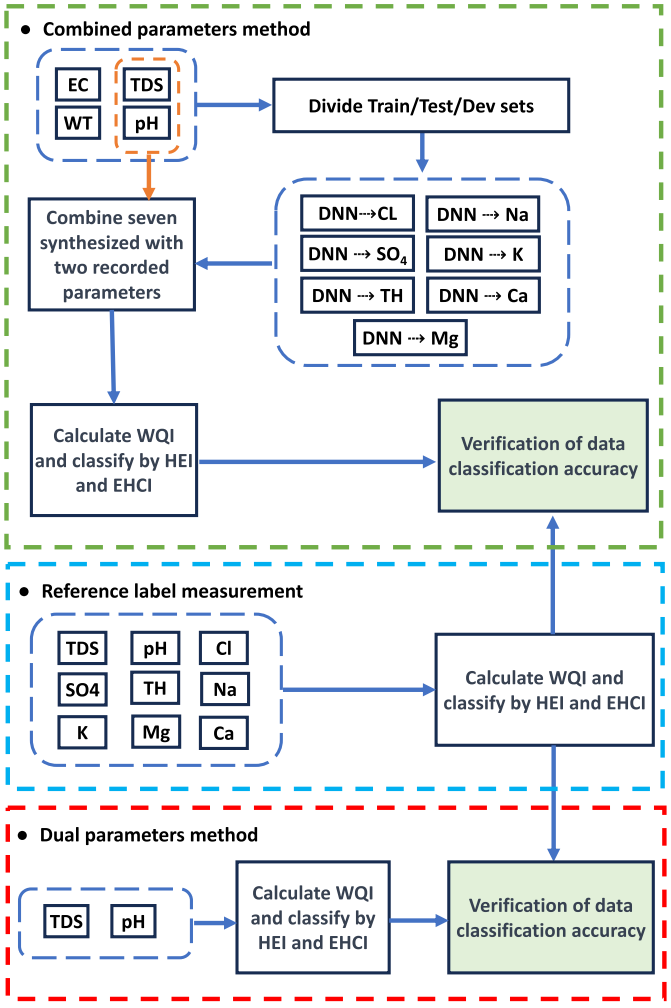


Fig. 1. Flow chart of the proposed water quality prediction system.

promote the formation and deposition of metal oxides and oxyhydrides, which can affect the water quality [35].

TH: The presence of calcium and magnesium in water contributes to its hardness. The aesthetic acceptability of drinking water to consumers is significantly influenced by its hardness level. Consumers are sensitive to changes in water hardness and can easily perceive them [34].

Ca: The presence of calcium in water can have both positive and negative impacts on taste. It can contribute to a desirable, slightly mineralized taste, enhancing palatability. However, excessive levels of calcium can result in a bitter or chalky taste, which may be undesirable to consumers [34].

Mg: Magnesium content in water can affect its taste. In moderate concentrations, it can enhance the flavor, giving a slightly sweet or smooth taste. However, excessive levels of magnesium can lead to an undesirable bitter or metallic taste [34]

SO_4: Elevated levels of sulfate in drinking water can have a discernible effect on taste, with high concentrations potentially causing a laxative effect in individuals unaccustomed to such levels [34].

K: The presence of potassium in drinking water is generally considered to be below health concern levels, and its impact on consumers is usually minimal unless they have specific conditions such as kidney dysfunction or heart diseases [34]. However, high levels of potassium in water can noticeably affect the taste, while lower levels of potassium tend to enhance the acceptance of water [36].

Na: The taste perception of sodium in water varies depending on factors such as the accompanying anion and the temperature of the water. Typically, at room temperature, the average concentration at which sodium becomes detectable by taste is around 200 mg/L [34].

TDS: The concentration of dissolved substances in water, known as TDS, is an essential parameter for assessing water quality. TDS includes salts, minerals, and organic compounds that are present in the water. When the TDS level is below approximately 600 mg/L, the water is typically considered palatable. However, if the TDS levels exceed about 1000 mg/L, the water becomes significantly unpalatable and can impact its taste and quality [34].

WT: Water temperature plays a significant role in determining its palatability. Generally, cool water is preferred over warm water due to its refreshing taste. Moreover, the temperature of water can impact the acceptability of various inorganic constituents and chemical contaminants that can affect its taste. Higher water temperatures can contribute to the growth of microorganisms and exacerbate issues related to taste, odor, color, and corrosion. The WHO does not provide any recommended permissible limit for water temperature [34].

2.1. Dataset

The Taiwan environmental protection administration (EPA) has diligently monitored groundwater at various sampling frequencies, including monthly, seasonal, and half-yearly intervals. The EPA regularly updates the monitoring data on their website, making it accessible to the public [37]. We utilized this dataset comprising 491 wells, which encompassed a total of 7702 water samples. Each sample in the dataset provided valuable information on 32 different water quality parameters. Notably, the dataset included essential spatial details such as the longitude and latitude coordinates of the sampling location, along with corresponding site numbers. Furthermore, temporal information, including the precise date and time of sample collection, was recorded. This extensive dataset covers the period from 2014 to 2022, offering a significant timeframe for conducting a thorough analysis and evaluation [37].

We performed a selection process to retain the most relevant water quality parameters for our analysis. The final dataset includes eleven key parameters: water temperature, electrical conductivity, potential of

hydrogen, total dissolved solids, magnesium, sulfate, potassium, sodium, calcium, total hardness, and chloride. In order to ensure data integrity, any water samples with missing values were excluded from the dataset. As a result, the filtered dataset comprises 4,845 water samples, each associated with the eleven selected parameters. Table 1 presents a statistical analysis of the filtered dataset, including the permissible limits set by the world health organization (WHO) for each parameter [34].

2.2. Water quality estimation

This study presents an innovative approach for predicting seven heavy metal parameters, including Mg, SO_4, K, Na, TH, Cl, and Ca, utilizing deep neural networks. Subsequently, water quality is assessed by integrating these parameters with recorded measurements of the TDS and pH. The approach leverages a concise set of four easily measurable parameters-WT, pH, EC, and TDS- to predict these seven essential heavy metal parameters. This holistic evaluation of water quality encompasses factors such as odor, taste, and appearance, aiming to enhance water quality analysis through the application of a deep neural network.

The utilization of the DNN enables us to effectively capture complex relationships and patterns in the data, enhancing the accuracy of our predictions. The DNN architecture consists of a fully connected five-layer network with neuron configurations of 32, 64, 512, 512, and 256 neurons in each respective layer (Fig. 2). Seven distinct DNNs were trained, each tailored to predict magnesium, sulfate, potassium, sodium, calcium, total hardness, and chloride levels. The input layer of these networks receives four essential parameters including temperature, electrical conductivity, potential of hydrogen, and total dissolved solids. Our method achieves accurate and consistent predictions by training dedicated models for each heavy metal parameter, leveraging a consistent architecture across all models. The hyperparameters of our DNN model are carefully chosen to optimize performance. We employ a learning rate of 0.001, a batch size of 32, and the ReLu activation function. Furthermore, batch normalization techniques are incorporated to enhance training stability and overall performance. The models for each heavy metal parameter are trained using the Adam optimizer and a specific number of epochs to achieve optimal convergence and accuracy. The maximum number of epochs for training the models is set at 1000. However, each individual model exhibits a different convergence point, with the convergence epochs for each parameter as follows: Mg (526), SO_4 (589), K (531), Na (309), TH (789), Cl (310), and Ca (257). This variation in convergence epochs underscores the unique learning patterns and complexities associated with each heavy metal parameter. Furthermore, considering the diverse ranges observed in Table 2 for the EC, pH, WT, and TDS parameters, we incorporate a crucial preprocessing step by normalizing the input data. This involves transforming the data to have a mean of zero and a standard deviation of one before passing it into our model. This normalization procedure enhances the DNN's ability to make accurate predictions by removing potential biases caused by varying scales of the input data. We establish uniform hyperparameter configurations for all models through systematic experimentation to ensure fair and consistent evaluation across all seven parameters. Our DNN model is trained using a standard approach of splitting the available data into three sets: training, validation, and testing. The training set comprises approximately 55 % of the data, while the remaining data is allocated to the validation and testing sets. The validation set accounts for 20 % of the data, and the testing set represents 25 % of the data.

Table 2
Performance of DNN model for both train/test data using R-squared (R^2), root mean square error (RMSE), and Pearson correlation coefficient (r).

Component	R^2		RMSE		r	
	Train	Test	Train	Test	Train	Test
Cl	0.98	0.98	240.02	300.42	1.00	1.00
SO_4	0.94	0.93	62.19	70.06	0.97	0.96
TH	0.97	0.97	109.83	107.36	0.99	0.99
Na	0.98	0.98	146.98	153.14	0.99	0.99
K	0.96	0.92	8.12	11.36	0.98	0.97
Mg	0.98	0.97	15.80	20.09	0.99	0.99
Ca	0.85	0.81	26.54	29.65	0.92	0.90

In the next step, we utilize seven artificially measured parameters (Mg, SO_4, K, Na, TH, Cl, and Ca) acquired through our method and two directly recorded parameters (TDS and pH) to compute the water quality index. We exclude the use of EC and temperature in the calculation of the water quality index, as the WHO does not provide permissible limits for these two parameters (refer to Table 1). We use heavy metal evaluation index [30,31] and entropy weighted heavy metal contamination index [32] approaches to assess the quality of each sample. These well-established methods are widely used for assessing the overall extent of water contamination and provide valuable insights for managing and preserving water resources.

2.2.1. Heavy metal evaluation index (HEI)

The heavy metal evaluation index is a commonly used method to assess the overall degree of contamination in water samples. The HEI values are calculated by summing up the contamination factors of individual parameters normalized by their maximum permissible values. The HEI values can be categorized into different pollution classes, including low (<10), moderate (10–20), and high (>20) [30,31].

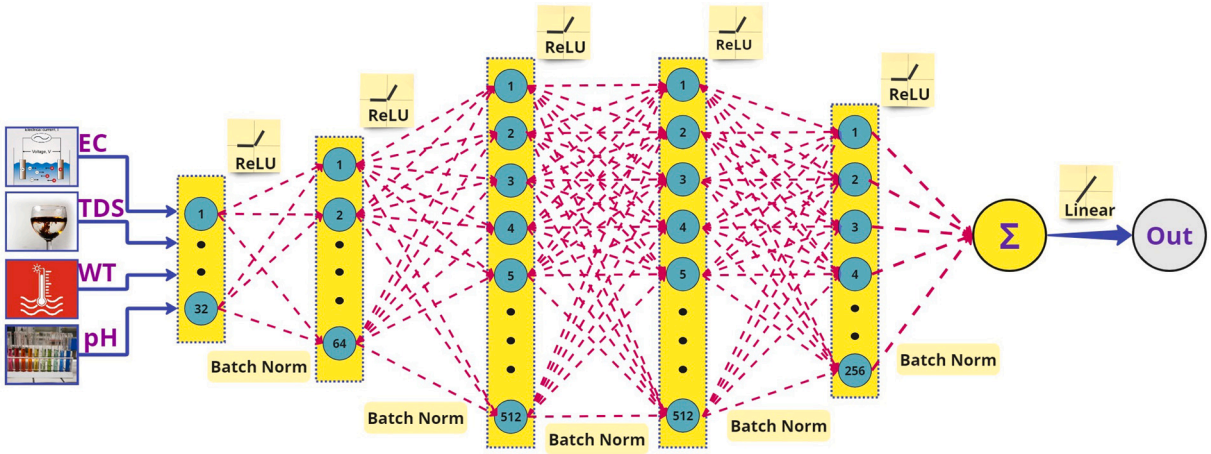


Fig. 2. Single deep neural network architecture employed across multiple instances for predicting seven parameters (Mg, SO_4, K, Na, TH, Cl, and Ca).

$$HEI = \sum_{i=1}^{n}\left(\frac{C_i}{S_i}\right) \quad (1)$$

where C_i is the concentration of the ith parameter (for instance Mg), S_i is the maximum permissible limit of the ith contaminant (for Mg is 250 mg/L), and n is the total number of parameters (nine parameters in our study including Mg, SO_4, K, Na, TH, Cl, Ca, pH, and TDS).

2.2.2. Entropy weighted heavy metal contamination index (EHCI)

The EHCI is a water quality index that leverages the concept of information entropy to assess the level of heavy metal contamination [32,33]. The method calculates an entropy weight (w_i) for each heavy metal using the Shannon entropy information technique [38]. The EHCI value is subsequently determined by summing the products of these weights and the corresponding sub-indices (Q_i). The EHCI values are classified into five categories based on their magnitude, with corresponding ranges as follows: excellent (<50), good (50–100), average (100–150), poor (150–200), and unsuitable (>200) [32].

The EHCI formula involves several parameters. The variable m represents the number of water samples analyzed, while n represents the number of parameters considered. The parameter C_i represents the concentration of the ith contaminant, while S_i represents the maximum permissible limit of the ith contaminant. v_{ij} is the normalized value of groundwater sample, and P_{ij} is the probability of the occurrence of the normalized value of the parameter j. The parameter E_j denotes the entropy of the jth heavy metal parameter, while w_j represents the entropy weight of the jth parameter. By incorporating these parameters, the EHCI formula provides a comprehensive assessment of heavy metal contamination in water samples, considering both the concentration of individual contaminants and their relative toxicity.

$$v_{ij} = \frac{C_{ij} - \min_m\{C_{mj}\}}{\max_m\{C_{mj}\} - \min_m\{C_{mj}\}} \quad (2)$$

$$P_{ij} = \frac{v_{ij}}{\sum_{i=1}^{m} v_{ij}} \quad (3)$$

$$E_j = -(1/lnm)\sum_{i=1}^{m} P_{ij} lnP_{ij} \quad (4)$$

$$w_j = \left(1 - E_j\right) \Big/ \sum_{i=1}^{m} 1 - E_j \quad (5)$$

$$Q_i = \left(\frac{C_i}{S_i}\right) * 100 \quad (6)$$

$$EHCI = \sum_{i=1}^{n} w_i Q_i \quad (7)$$

3. Results and discussion

In this section, we provide an extensive performance evaluation and analysis of the proposed approach, encompassing a range of metrics and experimental assessments. Firstly, we assess the accuracy and efficacy of the DNN model in predicting water quality parameters. Subsequently, we analyze the impact of using predicted values in understanding the quality of each water sample.

In assessing the performance of our model, we utilize essential metrics, namely R-squared (R^2), root mean square error (RMSE), and Pearson correlation coefficient (r). These metrics are applied on recorded and predicted parameters for both the training and testing datasets, providing a comprehensive assessment of the model's performance. A detailed summary of these metrics is presented in Table 2, offering an in-depth overview of the overall performance of the model. Table 2 reveals a strong correlation between the predicted and actual values, accompanied by high R-squared values. Moreover, the remarkably small root mean square error (RMSE) relative to the range of each parameter (refer to Table 2) serves as a strong indicator of the model's ability to make highly precise predictions for water quality parameters. These findings underscore the accuracy and reliability of our model in accurately predicting water quality parameters with a high level of precision. Fig. 3 offers a comparison between predicted values and actual values for the seven heavy metal parameters, encompassing the entire dataset. This visualization provides a holistic perspective on the model's performance in predicting these parameters, showcasing the consistency between the predicted and actual values and offering valuable insights into the overall accuracy and reliability of the predictive model for heavy metal parameters.

In our study, we utilized two water quality indices as the basis for classifying the water samples according to their quality. The classification process was conducted using two recorded parameters, specifically the pH and TDS. We refer to this classification approach as the “dual parameters” method, as it exclusively relies on these two recorded parameters for assigning water quality labels. Additionally, we expanded the parameter set by incorporating seven artificially synthesized parameters (Mg, SO_4, K, Na, TH, Cl, and Ca) alongside two recorded parameters pH and TDS. We refer to this extended approach as the “combined parameters” method, as it combines both the recorded and artificially synthesized parameters for classification.

In order to evaluate the impact of incorporating synthesized parameters on water sample classification, we employed nine actual recorded parameters as the *reference* standard. Consequently, we assign labels to each sample by considering these nine parameters, in conjunction with the two water quality indices, the HEI and EHCI. The distribution of samples across the HEI and EHCI classes reveals the following: in the HEI classification, the samples are categorized into three classes, namely safe (4349 samples), slightly polluted (293 samples), and highly polluted (203 samples). On the other hand, the EHCI classification divides the samples into five classes, including very safe (3903 samples), safe (534 samples), slightly polluted (85 samples), moderately polluted (45 samples), and highly polluted (273 samples). We conducted a comparison between the classification accuracy achieved by the dual parameters method and the combined parameters method, with the reference standard.

Fig. 4 illustrates the discrepancy between the dual parameters and combined parameters approaches across all classes. It clearly demonstrates that the dual parameters approach alone struggles to accurately identify water contamination, whereas the inclusion of predicted values for seven parameters in the combined parameters approach greatly enhances the prediction of all labels. Fig. 5 presents a comprehensive analysis of the classification performance for each individual class. It clearly demonstrates the consistent superiority of the combined feature approach over the dual parameters approach for almost all classes in both the EHCI and HEI contamination indices.

Regarding the HEI classification using the dual parameters approach, the results indicate that all slightly polluted samples (recall value 0) and approximately half of the highly polluted samples (recall value 0.52) are misclassified. However, the approach achieves perfect recall (recall value 1) for the safe class, indicating correctly classifying all samples with safe label. These findings highlight the limitations of the dual parameters approach in accurately classifying samples in these three classes. In contrast, the use of combined features outperforms the dual parameters approach, leading to improved classification performance. Similar trends are observed in the EHCI classification, where the combined parameters approach demonstrates higher accuracy and F1-score across all classes. Although there are slightly higher misclassification rates for highly polluted and moderately polluted samples compared to the dual parameters approach, the F1 scores for these classes highlight the superiority of using combined parameters. This indicates that

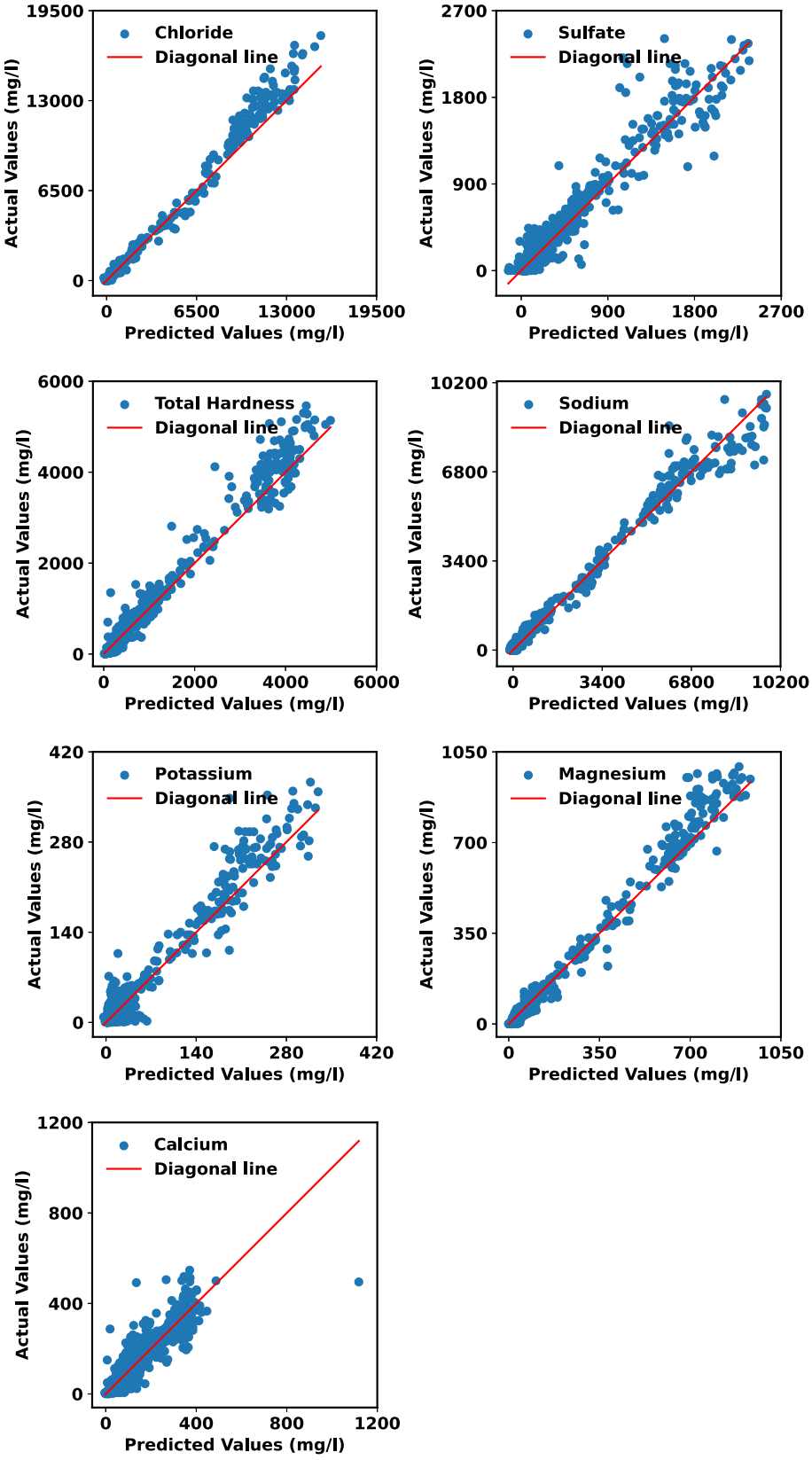


Fig. 3. Comparison between predicted and actual values of seven heavy metal parameters. The predicted values showcased in the figure were obtained through the utilization of a deep neural network as a regression method.

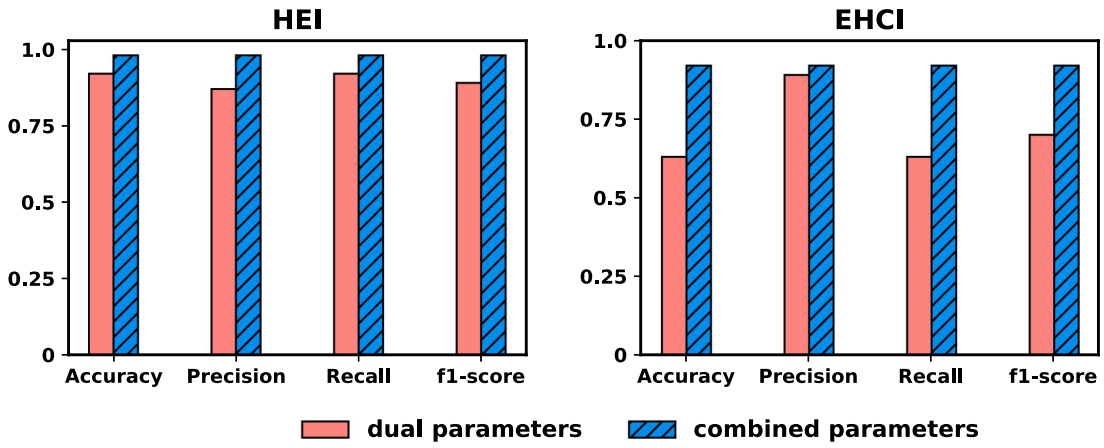


Fig. 4. General comparison between dual parameters and combined parameters approaches using the HEI and EHCI indices.

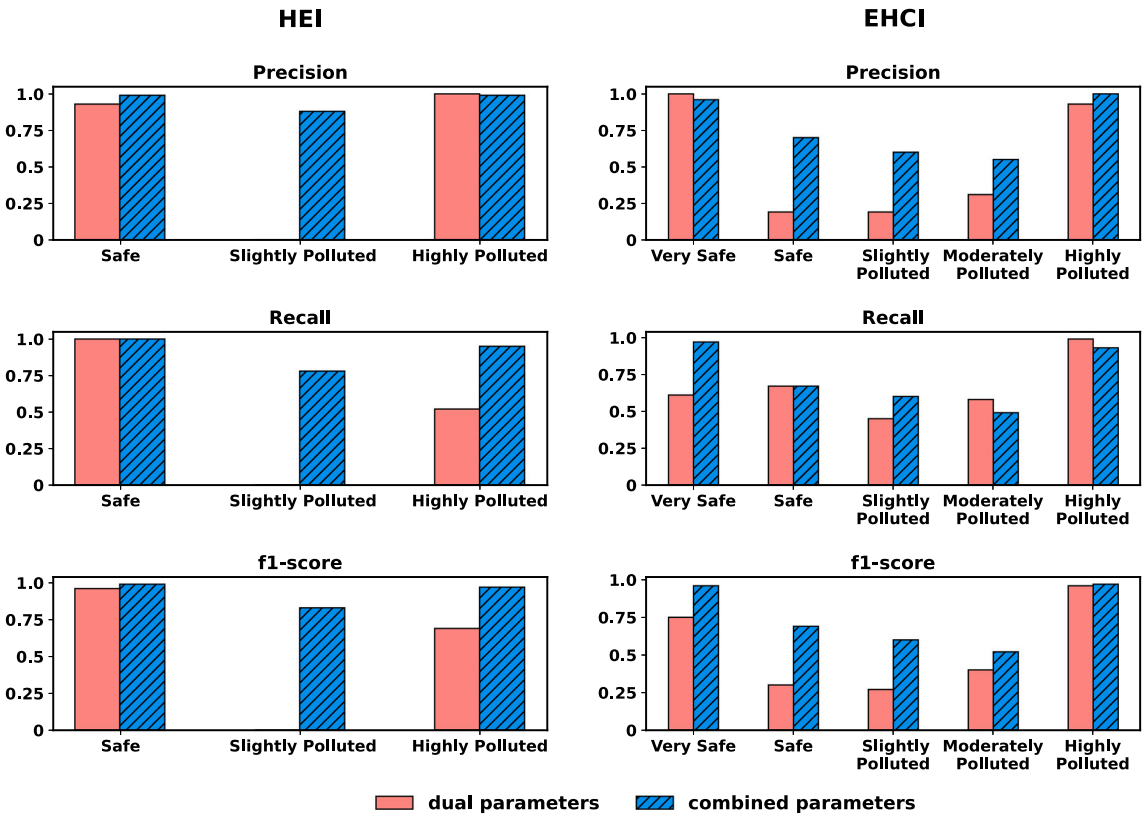


Fig. 5. Comparison of the dual parameters and combined parameters approaches for every individual classes using the HEI and EHCI indices.

employing the combined parameters approach is advantageous even for these specific classes. This detailed assessment of classification performance underscores the effectiveness and robustness of the combined feature approach in accurately identifying and differentiating various levels of contamination. The comparative analysis provides valuable insights into the performance and effectiveness of each approach in accurately classifying the water samples, while also considering the impact of synthesizing seven heavy metal parameters on our understanding of water quality.

Our approach excels in evaluating the palatability of water using only four easily measurable parameters, offering a cost-effective and accessible solution by virtually assessing seven heavy metal parameters. However, it is crucial to recognize the limitations of our approach in artificially measuring certain parameters such as Zinc (Zn), Lead (Pb), Cadmium (Cd), Chromium (Cr), and Copper (Cu). While these parameters may not directly affect the taste, odor, or color of water, they are highly relevant to assessing the safety of water. To depict these limitations, we utilize mutual information [38] a measure that quantifies the dependence or correlation between two random variables. Mutual information captures the shared information between variables based on their individual entropies and joint entropy. Our ability to accurately predict the aforementioned parameters is limited due to the low mutual information with temperature, EC, pH, and total dissolved solids, as depicted in Fig. 6. The high mutual information values between electrical conductivity and the parameters Pb, Cd, Cr, and Cu can be attributed to the discrete nature of these four parameters. Pb has only 18 unique values, Cd has 9 unique values, Cr has 13 unique values, and Cu has 18 unique values in the entire dataset. Mutual information

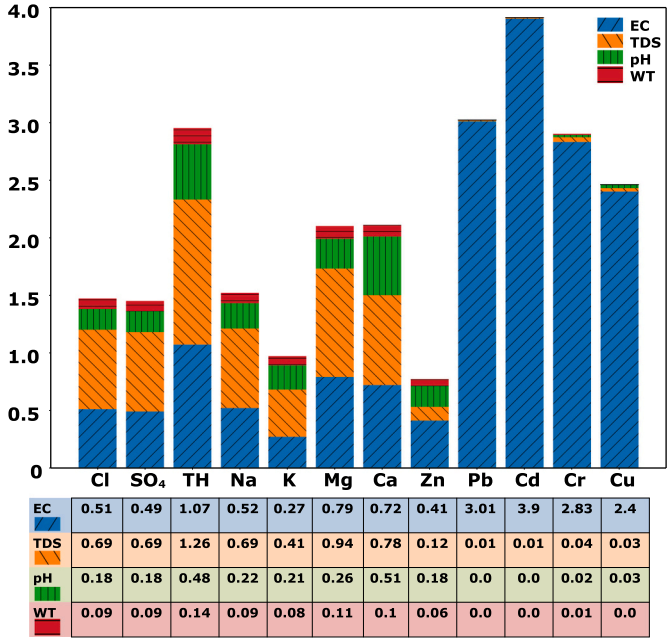


Fig. 6. Mutual information analysis between four initial parameters (temperature, EC, pH, and TDS) and twelve heavy metal parameters.

calculations assume a continuous distribution, which limits their effectiveness when applied to discrete data [39]. On the other hand, the predicted parameters (Mg, SO_4, K, Na, TH, Cl, and Ca) exhibit high mutual information with the four initial parameters. This indicates a strong correlation or dependence between these predicted parameters and the input parameters.

4. Conclusion

Water quality is a critical factor that directly influences its usability and acceptance among consumers. The taste, odor, and color of water are significant indicators of its quality and aesthetic acceptability. In this study, we successfully reconstructed seven heavy metal parameters (Mg, SO_4, K, Na, TH, Cl, and Ca) that impact these aesthetic characteristics using deep learning techniques. The water quality index is determined by evaluating seven artificially measured parameters in conjunction with two recorded ones. Samples are then classified into different water quality groups using the WQI value obtained from synthesized parameters. The WQI sample classification obtained from nine recorded parameters is used as a reference. This reference is employed to assess how well samples are classified by our method, which relies on synthesized parameters. We employed the entropy weighted heavy metal contamination index and the heavy metal evaluation index to evaluate the water quality. The results of our study reveal that the use of reconstructed parameters yields comparable performance to using actual recorded parameters. This suggests that the reconstructed parameters can effectively approximate the performance of actual recorded parameters. This research provides a cost-effective and efficient solution for assessing water quality, enabling better decision-making and effective management of water resources. Accurately estimating the water quality index using reconstructed parameters provides valuable insights into the taste, odor, and color characteristics of water samples.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Anusha Withana reports financial support was provided by Australian Research Council. Anusha Withana reports financial support was provided by Australia-India Strategic Research Fund.

Data availability

Data will be made available on request.

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